Akshay Kumar

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SUMMARY

Ph.D. in Mechanical Engineering (IIT Kanpur), specializing in FEA, vibration analysis, and experimental validation. Experienced in finite element modelling incorporating hyperelastic, viscoelastic, and nonlinear material behaviour, supported by ABAQUS simulations. I also have experience in machine learning and computer vision projects. My work has been published in peer-reviewed journal and conferences.

RESEARCH INTERESTS

Finite Element Analysis; Vibration Analysis; Solid Mechanics; Machine Learning

CORE SKILLS

FEA & Simulation: ABAQUS, HyperMesh, NASTRAN, PATRAN, COMSOL, AutoCAD, SolidWorks

Programming: Python, MATLAB

Experimental Characterization: Experimental Modal Analysis (EMA) using accelerometers, impact hammer, Digital Image Correlation (DIC) using high-speed cameras.

EDUCATION

Ph.D., Mechanical Engineering - Indian Institute of Technology Kanpur Aug 2020 - Dec 2026*

Joint M.Tech-Ph.D. Program · Specialization: Solid Mechanics & Design · CPI: 8.0/10

Dissertation: Experimental and Waveguide-Based Finite Element Investigations for Analyzing Vibrations in Tyres with Anisotropy, Hyperelasticity, Viscoelasticity, and Geometric Nonlinearity - Advisor: Prof. Nachiketa Tiwari

- Developed a nonlinear waveguide finite element (NWFEM) framework in MATLAB to model broadband vibration in radial tyres. Validated NWFEM predictions against ABAQUS and experimental data using modal testing (EMA).

*Expected PhD defence, Thesis submitted in July 2026

M.Tech, Mechanical Engineering - Indian Institute of Technology Kanpur Aug 2018 - Dec 2026
Specialization: Solid Mechanics and Design - CPI: 7.8/10

Indian Institute of Technology Kanpur, Kanpur, India

- Dissertation: Classification of Tabla Bols in Indian Music using Deep Learning Models
- Advisor: Prof. Nachiketa Tiwari

B.Tech, Mechanical Engineering - Meerut Institute of Engineering and Technology 2012 - 2016
71.4%

EXPERIENCE

FARE Fellow (Postdoctoral Research) - Tyre FEA and Vibration Sep 2026 - Present
Indian Institute of Technology Kanpur

- Currently extending the WFEM framework to incorporate air-cavity and tyre-rotation effects. Also working on a Physics-Informed Neural Network (PINN) for a vibration problem.

FEA Project - Submarine Project (sponsored by Mazagon Dock Shipbuilders Ltd.) March 2025 - May 2025
Indian Institute of Technology Kanpur · PI: Prof. Nachiketa Tiwari

- Delivered structural and vibration FEA modeling and simulation as part of a funded, timeline-bound R&D project

Machine Learning Intern March 2025 - Aug 2025

TSTS Pvt. Ltd., Kanpur

- Built computer vision and anomaly-detection models for railway applications

JOURNAL PUBLICATIONS

- **Akshay Kumar**, Madhavrao Londhe, Chirag Patel, Nachiketa Tiwari "An Experimentally Validated Waveguide-Based Finite Element Model for a Tyre with Hyperelastic, Viscoelastic and Anisotropic Effects." *Journal of Vibration Engineering & Technologies* (June 2026). DOI: 10.1007/s42417-026-02628-z

CONFERENCE PROCEEDINGS

- **Akshay Kumar**, Sreerag Ashok, Nachiketa Tiwari. "Indian music tabla bols classification using deep learning." 188th Meeting of the Acoustical Society of America, Louisiana, USA (May 2025). DOI: 10.1121/2.0002191
- Anubhav Mishra, **Akshay Kumar,** Nachiketa Tiwari, "Neural Network and Kinematic Control of a Stewart Platform for Animatronic Application," *6th International Conference on Recent Advances in Information Technology (RAIT)*, Dhanbad, India, 2025, DOI: 10.1109/RAIT65068.2025.11089023.

TEACHING EXPERIENCE

Tutor at IIT Kanpur

- Engineering Graphics (TA111, UG, 3 times) 2023–2026

Teaching Assistant at IIT Kanpur 2019–2022

- Engineering Acoustics and Its Control (ME778a, PG)
- Vibration & Control (ME354a, UG)

Teaching Assistant at NPTEL Swayam Courses 2022–2026

- Basics of Finite Element Analysis-I (3 times)
- Basics of Finite Element Analysis-II (3 times)
- Noise Management & Control
- Introduction to Composites

PROFESSIONAL WORK

1. Reviewer for journals: Journal of Vibration Engineering & Technologies (Springer, 3 times) 2026
2. Completed the Finlatics Data Science with Python Program Dec 2024 - Feb 2025
3. Conducted lectures on Signal Processing for HAL Employees under the Technical Leader Program at IIT Kanpur. 2023-2024
4. Attended four-week technical training in the Army Base workshop, Meerut, and NTPC, Delhi July 2015 - August 2015

POSITION OF RESPONSIBILITY

1. Senator, Student Gymkhana, IIT Kanpur 2019-2020
2. Sports Secretary, Hall 8, IIT Kanpur 2019-2020

REFERENCES

1. **Dr. Nachiketa Tiwari** (M.Tech - Ph.D. Thesis supervisor)
   Professor, Department of Mechanical Engineering,
   Indian Institute of Technology Kanpur, India
   Email: ntiwari@iitk.ac.in
2. **Dr. Akhilesh Mimani**
   Associate Professor, Department of Mechanical Engineering,
   Indian Institute of Technology Kanpur, India
   Email: amimani@iitk.ac.in
3. **Dr. Anubhav Mishra**
   Assistant Professor, Department of Design
   Indian Institute of Technology Guwahati, India
   Email: anubhv@iitg.ac.in